Visual vocabulary

Designing with data

There are so many ways to visualise data - how do we know which one to pick? Use the categories across the top to decide which data relationship is most important in your story, then look at the different types of chart within the category to form some initial ideas about what might work best. This list is not meant to be exhaustive, nor a wizard, but is a useful starting point for making informative and meaningful data visualisations.

FT graphic: Alan Smith; Chris Campbell; Ian Bott; Liz Faunce; Graham Parrish; Billy Ehrenberg-Shannon; Paul McCallum; Martin Stabe
Inspired by the Graphic Continuum by Jon Schwabish and Severino Ribecca

ft.com/vocabulary

Deviation

Emphasise variations (+/-) from a fixed reference point. Typically the reference point is zero but it can also be a target or a long-term average. Can also be used to show sentiment (positive/neutral/negative).

Example FT uses
Trade surplus/deficit, climate change

Diverging bar



A simple standard bar chart that can handle both negative and positive magnitude values.

Diverging stacked bar



Perfect for presenting survey results which involve sentiment (eg disagree/neutral/agree).

Spine



Splits a single value into two contrasting components (eg male/female).

Surplus/deficit filled line



The shaded area of these charts allows a balance to be shown – either against a baseline or between two series.

Correlation

Show the relationship between two or more variables. Be mindful that, unless you tell them otherwise, many readers will assume the relationships you show them to be causal (i.e. one causes the other).

Example FT uses
Inflation and unemployment, income and life expectancy

Scatterplot



The standard way to show the relationship between two continuous variables, each of which has its own axis.

Column + line timeline



A good way of showing the relationship between an amount (columns) and a rate (line).

Connected scatterplot



Usually used to show how the relationship between 2 variables has changed over time.

Bubble



Like a scatterplot, but adds additional detail by sizing the circles according to a third variable.

XY heatmap



A good way of showing the patterns between 2 categories of data, less effective at showing fine differences in amounts.

Ranking

Use where an item's position in an ordered list is more important than its absolute or relative value. Don't be afraid to highlight the points of interest.

Example FT uses
Wealth, deprivation, league tables, constituency election results

Ordered bar



Standard bar charts display the ranks of values much more easily when sorted into order.

Ordered column



See above.

Ordered proportional symbol



Use when there are big variations between values and/or seeing fine differences between data is not so important.

Dot strip plot



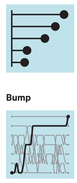
Dots placed in order on a strip are a space-efficient method of laying out ranks across multiple categories.

Slope



Perfect for showing how ranks have changed over time or vary between categories.

Lollipop



Lollipops draw more attention to the data value than standard bar/column and can also show rank and value effectively.

Bump

Effective for showing changing rankings across multiple dates. For large datasets, consider grouping lines using colour.

Distribution

Show values in a dataset and how often they occur. The shape (or 'skew') of a distribution can be a memorable way of highlighting the lack of uniformity or equality in the data.

Example FT uses
Income distribution, population (age/sex) distribution, revealing inequality

Histogram



The standard way to show a statistical distribution - keep the gaps between columns small to highlight the 'shape' of the data.

Dot plot



A simple way of showing the change or range (min/max) of data across multiple categories.

Dot strip plot



Good for showing individual values in a distribution, can be a problem when too many dots have the same value.

Barcode plot



Like dot strip plots, good for displaying all the data in a table, they work best when highlighting individual values.

Boxplot



Summarise multiple distributions by showing the median (centre) and range of the data

Violin plot



Similar to a box plot but more effective with complex distributions (data that cannot be summarised with simple average).

Population pyramid



A standard way for showing the age and sex breakdown of a population distribution; effectively, back to back histograms.

Cumulative curve



A good way of showing how unequal a distribution is: y axis is always cumulative frequency, x axis is always a measure.

Frequency polygons



For displaying multiple distributions of data. Like a regular line chart, best limited to a maximum of 3 or 4 datasets.

Beeswarm



Use to emphasise individual points in a distribution. Points can be sized to an additional variable. Best with medium-sized datasets

Change over Time

Give emphasis to changing trends. These can be short (intra-day) movements or extended series traversing decades or centuries: Choosing the correct time period is important to provide suitable context for the reader.

Example FT uses
Share price movements, economic time series, sectoral changes in a market

Line



The standard way to show a changing time series. If data are irregular, consider markers to represent data points.

Column



Columns work well for showing change over time - but usually best with only one series of data at a time.

Column + line timeline



A good way of showing the relationship over time between an amount (columns) and a rate (line).

Slope



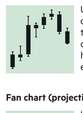
Good for showing changing data as long as the data can be simplified into 2 or 3 points without missing a key part of story.

Area chart



Use with care – these are good at showing changes to total, but seeing change in components can be very difficult.

Candlestick



Usually focused on day-to-day activity, these charts show opening/closing and high/low points of each day.

Fan chart (projections)



Use to show the uncertainty in future projections - usually this grows the further forward to projection.

Connected scatterplot



A good way of showing changing data for two variables whenever there is a relatively clear pattern of progression.

Calendar heatmap



A great way of showing temporal patterns (daily, weekly, monthly) – at the expense of showing precision in quantity.

Priestley timeline



Great when date and duration are key elements of the story in the data.

Circle timeline



Good for showing discrete values of varying size across multiple categories (eg earthquakes by continent).

Vertical timeline



Presents time on the Y axis. Good for displaying detailed time series that work especially well when scrolling on mobile.

Seismogram



Another alternative to the circle timeline for showing series where there are big variations in the data.

Streamgraph



A type of area chart; use when seeing changes in proportions over time is more important than individual values

Magnitude

Show size comparisons. These can be relative (just being able to see larger/bigger) or absolute (need to see fine differences). Usually these show a 'counted' number (for example, barrels, dollars or people) rather than a calculated rate or per cent.

Example FT uses
Commodity production, market capitalisation, volumes in general

Column



The standard way to compare the size of things. Must always start at 0 on the axis.

Bar



See above. Good when the data are not time series and labels have long category names.

Paired column



As per standard column but allows for multiple series. Can become tricky to read with more than 2 series.

Paired bar



See above.

Marimekko



A good way of showing the size and proportion of data at the same time – as long as the data are not too complicated.

Proportional symbol



Use when there are big variations between values and/or seeing fine differences between data is not so important.

Isotype (pictogram)



Excellent solution in some instances – use only with whole numbers (do not slice off an arm to represent a decimal).

Lollipop



Lollipop charts draw more attention to the data value than standard bar/column – does not have to start at zero (but preferable).

Radar



A space-efficient way of showing value of multiple variables– but make sure they are organised in a way that makes sense to reader.

Parallel coordinates



An alternative to radar charts – again, the arrangement of the variables is important. Usually benefits from highlighting values.

Bullet



Good for showing a measurement against the context of a target or performance range.

Grouped symbol



An alternative to bar/column charts when being able to count data or highlight individual elements is useful.

Part-to-whole

Show how a single entity can be broken down into its component elements. If the reader's interest is solely in the size of the components, consider a magnitude-type chart instead.

Example FT uses
Fiscal budgets, company structures, national election results

Stacked column/bar



A simple way of showing part-to-whole relationships but can be difficult to read with more than a few components.

Marimekko



A good way of showing the size and proportion of data at the same time – as long as the data are not too complicated.

Pie



A common way of showing part-to-whole data – but be aware that it's difficult to accurately compare the size of the segments.

Donut



Similar to a pie chart – but the centre can be a good way of making space to include more information about the data (eg total).

Treemap



Use for hierarchical part-to-whole relationships; can be difficult to read when there are many small segments.

Voronoi



A way of turning points into areas – any point within each area is closer to the central point than any other centroid.

Arc



A hemicycle, often used for visualising parliamentary composition by number of seats.

Gridplot



Good for showing % information, they work best when used on whole numbers and work well in small multiple layout form.

Venn



Generally only used for schematic representation.

Waterfall



Can be useful for showing part-to-whole relationships where some of the components are negative.

Spatial

Aside from locator maps only used when precise locations or geographical patterns in data are more important to the reader than anything else.

Example FT uses
Population density, natural resource locations, natural disaster risk/impact, catchment areas, variation in election results

Basic choropleth (rate/ratio)



The standard approach for putting data on a map – should always be rates rather than totals and use a sensible base geography.

Proportional symbol (count/magnitude)



Use for totals rather than rates – be wary that small differences in data will be hard to see.

Flow map



For showing unambiguous movement across a map.

Contour map



For showing areas of equal value on a map. Can use deviation colour schemes for showing +/- values

Equalised cartogram



Converting each unit on a map to a regular and equally-sized shape – good for representing voting regions with equal value.

Scaled cartogram (value)



Stretching and shrinking a map so that each area is sized according to a particular value.

Dot density



Used to show the location of individual events/locations – make sure to annotate any patterns the reader should see.

Heat map



Grid-based data values mapped with an intensity colour scale. As choropleth map – but not snapped to an admin/political unit.

Flow

Show the reader volumes or intensity of movement between two or more states or conditions. These might be logical sequences or geographical locations.

Example FT uses
Movement of funds, trade, migrants, lawsuits, information; relationship graphs.

Sankey



Shows changes in flows from one condition to at least one other; good for tracing the eventual outcome of a complex process.

Waterfall



Designed to show the sequencing of data through a flow process, typically budgets. Can include +/- components.

Chord



A complex but powerful diagram which can illustrate 2-way flows (and net winner) in a matrix.

Network



Used for showing the strength and inter-connectedness of relationships of varying types.

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